# Vusumuzi Nkosi

Data Scientist · Forecasting · Fraud & Risk Analytics · Experimentation

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## SUMMARY

Data scientist specialising in **forecasting, fraud and risk analytics, and experimentation**. I start from the decision a team has to make, prove each model beats a simple baseline, and explain the result in plain language, a skill sharpened over nine years of teaching mathematics. Recent work: **3.1% error** forecasting South Africa's electricity demand a week ahead (and 2.4% a day ahead, against Eskom's own published 2.3%), **98% of fraud caught** within a team's daily review capacity, and **82% of revenue at risk protected** by a ranked churn call list. Completing a PGDip in Data Science (STADIO, NQF 8); seeking data scientist or data analyst roles in South Africa or remote.

## TECHNICAL SKILLS

**Programming:** Python, SQL (CTEs, window functions, DuckDB), R (basic)

**ML & Modelling:** scikit-learn, XGBoost, gradient boosting, random forest, logistic regression, K-means clustering, feature engineering, cost-based thresholds, time-based and walk-forward validation, imbalanced data

**Statistics & Experimentation:** hypothesis testing, A/B testing, McNemar's test, confidence intervals, power analysis, regression analysis, time-series forecasting, SciPy, statsmodels

**Data & Visualisation:** pandas, NumPy, Matplotlib, Seaborn, Streamlit, Tableau, Excel

**Tools & Platforms:** Git, GitHub, Jupyter Notebook, Google Colab, AWS SageMaker, Prompt Engineering, Generative AI

## PROJECTS

**Eskom Demand Forecasting** — Python, scikit-learn, gradient boosting

*github.com/vusumuzinkosi/eskom-demand-forecasting*

- Forecast South Africa's hourly electricity demand a week ahead from 4.5 years of Eskom data (39,600 hours), using calendar, public-holiday and lagged-demand features with 52 weeks of walk-forward validation.
- Reached 3.1% MAPE a week ahead, 15% better than a “same as last week” baseline, and 2.4% a day ahead against Eskom's published 2.3%, without weather data.

**Fraud Detection: A Daily Review Queue for Mobile Money** — Python, scikit-learn

*github.com/vusumuzinkosi/fraud-detection-mobile-money*

- Built a daily review queue for a six-analyst team from 2.6 million mobile money transactions, with a time-based split and only information known at transaction time; found and removed a leaking balance column.
- Reviewing the top 300 transactions a day catches 98% of fraud cases (89% of the money) with 87% of flags genuine, where the existing rule caught 1 fraud in 4 days.

**Churn Prediction: Revenue at Risk** — Python, scikit-learn, Streamlit

*github.com/vusumuzinkosi/churn-prediction-revenue-at-risk · Live demo: vusumuzinkosi-churn-planner.streamlit.app*

- Trained logistic regression and random forest models with a decision threshold set from the business costs of each error; the random forest catches 95% of churners on unseen customers (ROC AUC 0.98).
- Ranking customers by expected loss protects 82% of the value at risk at 10% contact capacity, against 58% when ranking by churn probability; exported a ranked call list with a spending ceiling per customer.
- Built and deployed a Streamlit planner where a retention manager sets capacity, costs and save rate and gets the break-even call list, value protected and expected net value, using out-of-fold scores for all 2,850 customers.

**A/B Test: New Landing Page** — Python, statsmodels, SciPy

*github.com/vusumuzinkosi/ab-test-landing-page*

- Pre-registered a two-sided test (α = 0.05, +0.5-point minimum worthwhile lift) on 290,584 visitors, after cleaning mismatched rows and duplicate users.
- Recommended not rolling out: the new page converted 0.16 points lower (p = 0.19, 95% CI −0.39 to +0.08), with 98% power for the lift that matters.

**Customer Segmentation: RFM and K-means** Python, SQL (DuckDB), scikit-learn

*github.com/vusumuzinkosi/customer-segmentation-rfm*

- Segmented 5,852 customers from two years of retail transactions with hand-built RFM tiers, then K-means on log-scaled RFM with k chosen by silhouette score and business use.
- Found loyal big spenders are 19% of customers but 72% of revenue; segments built on older data predicted the next six months, with the top group spending 19× as much as lapsed buyers.
- Rebuilt the cleaning and RFM pipeline in SQL with DuckDB (CTEs, window functions, cohort retention), matching pandas exactly on 776,577 lines; found only about 1 in 5 new customers reorders within a month.

## WORK EXPERIENCE

**Senior Lecturer, Mathematics & IT** Feb 2017 – Present

*Amajuba TVET College — Newcastle, KwaZulu-Natal*

- Analyse learner performance data to identify at-risk students early and target evidence-based interventions, improving pass rates.
- Teach mathematics and IT to 100+ students a year, designing assessments and curriculum delivery aligned with national standards.
- Explain quantitative concepts to non-specialist audiences every day, the same skill used to present model results to business stakeholders.
- Developed standardised assessment rubrics adopted department-wide, and mentored IT competition teams to national rounds.

**Special Educator, Mathematics** Jan 2012 – Mar 2016

*Mdumiseni High School — KwaZulu-Natal*

- Taught mathematics to diverse learner groups, tracking individual progress and adapting materials to different learning needs.

## EDUCATION

**Postgraduate Diploma in Data Science (NQF 8)** 2026 (in progress)

*STADIO Higher Education — Remote*

- 120-credit qualification (SAQA ID: 122721); modules: Introduction to Data Science, Working with Data, Statistical Modelling, Applied Machine Learning, Capstone Project.

**Data Science Programme** Mar 2025 – Apr 2026

*ALX Africa — Remote*

**Postgraduate Diploma in TVET** 2024 – 2025

*University of the Western Cape*

**BSc Mathematics & Computer Science** 2019 – 2024

*University of South Africa*

**BEd Mathematics & Computer Science** 2012 – 2017

*University of South Africa*

## CERTIFICATIONS

- Data Science Certificate — ALX Africa (2025–2026)
- Data Analyst Certificate — ALX Africa (2025): SQL, Power BI, A/B Testing, Regression Analysis
- AI Engineering Bootcamp — Zero To Mastery Academy (2025)
- Prompt Engineering Bootcamp — Zero To Mastery Academy (2025)